

AARHUS UNIVERSITY

DEPARTMENT OF MATHEMATICS

INTRODUCTION TO ALGEBRA 1

Handin 5

Author

Søren M. DAMSGAARD

Student number

202309814

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Goofy ah Math Number 5

Digit sum!

Let's remind ourselves what the digit sum is, so look at 12345 that has the digit sum $1 + 2 + 3 + 4 + 5 = 15$ and generally the a number can be represented as

$$a_n a_{n-1} \dots a_1 a_0 = \sum_{i=0}^n a_i 10^i \quad (1)$$

where the left-side represent the digits of the number, not some operation between some numbers $a_n a_{n-1} \dots a_1 a_0$

Show some $n \in \mathbb{N}$ is $3|n$ iff. the digit sum of n , n_{tsum} is also $3|n_{tsum}$

This is equivalent to show

$$n_{tsum} \equiv 0 \pmod{3} \Leftrightarrow n \equiv 0 \pmod{3}$$

start assuming that $n_{tsum} \equiv 0 \pmod{3}$ and rewriting n_{tsum} using (1) and let

$$n_{tsum} = a_n + a_{n-1} + \dots + a_1 + a_0 \quad (2)$$

the substitute n_{tsum} with the above

$$a_n + a_{n-1} + \dots + a_1 + a_0 \equiv 0 \pmod{3} \quad (3)$$

Then we want to show for all $a \in n_{Tsum}$ that $a \cdot 10^i \equiv a \cdot 10^k \pmod{3}$.

To do that let's look at properties of the number 10

$$10 \equiv 1 \pmod{3}$$

then remember that multiplication is preserved in congruence, so

$$10^i \equiv 1 \pmod{3} \text{ for } i \in \mathbb{N}$$

and then use lemma 9.2 [1] again, so

$$\begin{aligned} a_i \cdot 10^i &\equiv a_i \cdot 1 \pmod{3} \\ &= a_i \cdot 10^i \equiv a_i \pmod{3} \end{aligned}$$

then apply this to every factor in n_{Tsum} to get

$$\begin{aligned} a_n \cdot 10^n &\equiv a_n \pmod{3} \\ a_{n-1} \cdot 10^{n-1} &\equiv a_{n-1} \pmod{3} \\ &\vdots \\ a_1 \cdot 10^1 &\equiv a_1 \pmod{3} \\ a_0 \cdot 10^0 &\equiv a_0 \pmod{3} \end{aligned}$$

then use lemma 9.2 [1] again, to construct n

$$\sum_{i=0}^n a_i 10^i \equiv \sum_{i=0}^n a_i \cdot 10^i \pmod{3}$$

which is

$$\begin{aligned} n &\equiv a_n + a_{n-1} + \dots + a_1 + a_0 \pmod{3} \\ n &\equiv n_{Tsum} \pmod{3} \end{aligned}$$

and remembering the original assumption $n_{Tsum} \equiv 0 \pmod{3}$ leads to the following

$$n \equiv 0 \pmod{3} \tag{4}$$

This can also be shown the other way around

Show some $n \in \mathbb{N}$ is $9|n$ iff. the digit sum of n , n_{tsum} is also $9|n$

The proof for this is equivalent to the above where you replace modulo 3 with modulo 9, since it holds that

$$10 \equiv 1 \pmod{9}$$

so using lemma 9.2 [1] results in the same conclusion.

let $n \in \mathbb{N}$ and let m be a number that occurs when you shift around the individual numbers in n . Show that $n - m$ is a multiple of 9

Let n_{Tsum} be the digit sum of n and let m_{Tsum} be the digit sum of m . Since m is formed by jumbling the digits of n the digit sum of m and n is the same

$$n_{Tsum} = m_{Tsum}$$

and it's shown from the first part that a number n is always congruent to its digit sum modulo 9, this gives the following

$$\begin{aligned} n &\equiv n_{Tsum} \pmod{9} \\ m &\equiv m_{Tsum} \pmod{9} \end{aligned}$$

and using the fact that their digit sums are the same

$$n \equiv m \pmod{9}$$

then using lemma 9.2. [1] to subtract m from both sides

$$n - m \equiv 0 \pmod{9}$$

which means the sum $n - m$ is a multiple of 9.

References

- [1] Johan P. Hansen. *Tal og polynomier. Begreber, metoder, resultater, kodning og kryptografi*. Aarhus Universitetsforlag, 2018.